

Matthew Buchholz

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PERSONAL STATEMENT

I study optimization and training dynamics in deep learning, with the goal of using theory to predict and improve empirical behavior in modern neural networks. My work focuses on stochastic optimization methods, feature learning, and training stability, combining mathematical analysis with empirical validation on large-scale vision and language models.

EDUCATION

- University of British Columbia** 2025 – curr.
Ph.D. Candidate in Computer Science
Advised by Dr. Mark Schmidt, Dr. Danica J. Sutherland, and Dr. Geoff Pleiss
GPA: 4.0
- New York University:** Courant Institute of Mathematical Sciences 2022 – 2025
M.Sc. in Mathematics
Thesis: *On Approximation by Shallow Neural Networks*
Advised by Dr. Sinan Güntürk and Dr. Joan Bruna
- California State University, Long Beach** 2018 – 2021
B.Sc. Computer Science; B.A. Physics; Minor in Mathematics
Advised by Dr. Wenlu Zhang

RESEARCH EXPERIENCE

- Convergence Analysis of Schedule-Free SGD** May 2025 – Present
Collaborators: Robert Gower and Mandy Pun Flatiron Institute
- Derived convergence guarantees for schedule-free SGD for convex objectives, extending the theory for arbitrary learning-rate schedules.
 - Ran PyTorch experiments on vision and language modeling tasks comparing SGD with momentum, Polyak-Ruppert Averaging, and schedule-free variants, demonstrating close agreement between theory and observed training dynamics.
- Approximation Theory of Deep Neural Networks** Sep 2023 – May 2025
Advisors: Dr. Sinan Güntürk and Dr. Joan Bruna New York University
- Studied approximation properties of shallow and deep neural networks, including depth separations and infinite-width limits.
- Recurrent Attentive Kernel Learning** Jan 2021 – May 2021
Advisor: Dr. Wenlu Zhang California State University, Long Beach
- Developed a novel deep kernel learning architecture combining attention weighted RNN features with Gaussian process predictions via type-II maximum likelihood for time-series modeling.
 - Achieved 94.5% test accuracy (0.95 F1 metric) on shark behavioral classification from accelerometer data (CSULB Shark Lab collaboration), outperforming baseline time-series models using feature extraction DTW, wavelets, stationarity decompositions by 5%.
 - Paper accepted to KDD 2021 Workshop on Mining and Learning from Time Series.

Deep Learning for Additive Manufacturing

Advisor: Dr. Wenlu Zhang

Aug 2020 – May 2021

California State University, Long Beach

- Applied functional principal component analysis (FPCA) and transfer learning from natural image datasets to predict porosity from infrared thermal images of the melt pool in additive manufacturing
- Paper accepted as journal paper in IEEE Access.

WORK EXPERIENCE

Guest Researcher

May 2025 – Sep 2025

Flatiron Institute Center for Computational Mathematics

New York, NY

- Implemented large-scale benchmarks comparing optimization algorithms (momentum SGD, Polyak-Ruppert averaging, Adam, schedule-free variants) across vision and language modeling tasks

Machine Learning Engineer

Sep 2021 – Jun 2023

Samsung Research America

Mountain View, CA

Worked on the Think Tank Team (TTT), a multi-disciplinary R&D team, fast prototyping machine learning solutions, research software and hardware development, design, and product development

- Fined tuned vision transformer (ViT) model for representation learning on video, using dimensionality reduction in latent space for anomaly detection.
- Implemented spherical convolutional neural networks with optimal transport loss function for extracting lighting from smart phone images.
- Implemented kernel regression on raw audio data for temperature prediction.

Machine Learning Engineer Intern

Jun 2021 – Sep 2021

Samsung Research America

Mountain View, CA

- Designed a generative convolutional neural network with spectral regularization for generating inverse image filters to remove hardware specific blur/diffraction.
- Implemented convex optimization baseline using proximal gradient methods.

SDE Intern

Jun 2020 – Aug 2020

Amazon

Seattle, WA

- Developed cross-platform (iOS + Android) features for Kindle that improved the client's efficiency within lower-level rendering system (C++)

Software Developer Intern

Jun 2019 – Aug 2019

The Aerospace Corporation

El Segundo, CA

Application Development Intern

May 2018 – Sep 2018

Siemens PLM Software

Cypress, CA

PUBLICATIONS

- [1] Y.-M. Pun, M. Buchholz, and R. M. Gower, *Schedulers for schedule-free: Theoretically inspired hyperparameters*, (Under Review), 2025. arXiv: 2511.07767 [cs.LG].
- [2] M. Buchholz, W. Zhang, E. N. Meese, Y. Yang, C. G. Lowe, and H.-G. Yeh, "Recurrent attentive kernel learning for shark activity recognition", *7th SIGKDD Workshop on Mining and Learning from Time Series (MiLeTS)*, 2021.
- [3] S. Ho, W. Zhang, W. Young, M. Buchholz, S. Al Jufout, K. Dajani, L. Bian, and M. Mozumdar, "Dlam: Deep learning based real-time porosity prediction for additive manufacturing using thermal images of the melt pool", *IEEE Access*, vol. 9, pp. 115 100–115 114, 2021.

SKILLS

- **Languages:** Python, C++, Julia, SQL
- **ML Frameworks:** PyTorch, JAX, Pyro, GPyTorch, OpenCV, ROS
- **Research Areas:** Deep Learning Theory, Optimization, Gaussian Processes, Generative Models, Inverse Problems

HONORS AND AWARDS

- **President's Academic Excellence Initiative PhD Award:** Winter 2025
- **President's list:** Spring 2018, Fall 2018, Spring 2019, Spring 2020
- **Dean's list:** Fall 2019
- **Tau Beta Pi Engineering Honor Society** (Top 1/8th of Junior Class - CSULB Chapter)

SELECTED TALKS

SIGKDD Workshop on Mining and Learning from Time Series (MiLeTS), Virtual <i>Recurrent Attentive Kernel Learning for Shark Activity Recognition</i>	Aug 2021
Long Beach Deep Learning Lab, Long Beach, CA <i>Bayesian Machine Learning</i>	Mar 2021
CATIE Research Group, Long Beach, CA <i>Functional Principal Component Analysis</i>	Sep 2020
The Aerospace Corporation, El Segundo, CA <i>Reproducing Kernel Hilbert Space</i>	Aug 2019
Long Beach Deep Learning Lab, Long Beach, CA <i>Gaussian Visual-Semantic Embedding for Joint Image-Text Representation</i>	Jun 2019

REVIEWING

- SIGKDD 2020